

Boss Challenge — Paper 19 (Class 7)

Motion & Time | Heat | Fractions & Decimals | Perimeter & Area

One correct option each. Marking: +4 correct, -1 wrong, 0 blank. Write A/B/C/D on a separate sheet.

1. The distance a body covers divided by the time it takes is what we call its:
(A) speed
(B) distance
(C) weight
(D) time
2. The sum $\frac{1}{2} + \frac{1}{4}$ equals:
(A) $\frac{3}{4}$
(B) $\frac{2}{6}$
(C) $\frac{1}{6}$
(D) $\frac{1}{2}$
3. Heat always flows on its own from a body at:
(A) lower temperature to one at higher temperature
(B) a larger body to a smaller body
(C) higher temperature to one at lower temperature
(D) a heavier body to a lighter body
4. The perimeter of a rectangle of length l and breadth b is given by:
(A) $l \times b$
(B) $2 \times (l + b)$
(C) $4 \times l$
(D) $l + b$
5. In the SI system, the unit used to measure speed is the:
(A) second (s)
(B) metre (m)
(C) kilogram (kg)
(D) metre per second (m/s)
6. Multiplying the fractions $\frac{2}{3}$ and $\frac{3}{4}$ together gives a result of:
(A) $\frac{6}{7}$
(B) $\frac{1}{2}$
(C) $\frac{5}{7}$
(D) $\frac{5}{12}$
7. The quantity that tells us how hot or cold a body is, measured with a thermometer, is its:
(A) temperature
(B) weight
(C) length
(D) speed

8. The area of a rectangle of length l and breadth b is:
- (A) $l \times b$
 - (B) $l + b$
 - (C) $4 \times l$
 - (D) $2 \times (l + b)$
9. If a car covers a distance of 60 km in 2 hours, then its speed works out to:
- (A) 62 km/h
 - (B) 120 km/h
 - (C) 30 km/h
 - (D) 15 km/h
10. Dividing $\frac{1}{2}$ by $\frac{1}{4}$ gives the value:
- (A) $\frac{1}{2}$
 - (B) $\frac{1}{8}$
 - (C) 2
 - (D) 8
11. To measure the temperature of the human body, a doctor uses a:
- (A) clinical thermometer
 - (B) speedometer
 - (C) laboratory thermometer
 - (D) barometer
12. A square has a side of 5 cm. Its perimeter is:
- (A) 25 cm
 - (B) 10 cm
 - (C) 9 cm
 - (D) 20 cm
13. When a body covers equal distances in equal intervals of time, its motion is described as:
- (A) non-uniform motion
 - (B) circular motion
 - (C) uniform motion
 - (D) no motion at all
14. The sum $0.5 + 0.25$ equals:
- (A) 7.5
 - (B) 0.7
 - (C) 0.75
 - (D) 0.30
15. A healthy person's body stays at a normal temperature of roughly:
- (A) 50°C
 - (B) 100°C
 - (C) 37°C
 - (D) 0°C

16. A square has a side of 5 cm. Its area is:

- (A) 10 cm^2
- (B) 55 cm^2
- (C) 20 cm^2
- (D) 25 cm^2

17. A bus that speeds up, slows down and speeds up again as it moves through traffic is showing:

- (A) uniform motion
- (B) periodic motion
- (C) rest
- (D) non-uniform motion

18. Multiplying the decimals 0.2 and 0.3 gives:

- (A) 0.6
- (B) 0.06
- (C) 6.0
- (D) 0.5

19. Heat passing through a solid metal rod, from its hot end to its cold end, travels by:

- (A) reflection
- (B) convection
- (C) conduction
- (D) radiation

20. A rectangle is 8 cm long and 3 cm broad. Its area is:

- (A) 24 cm^2
- (B) 22 cm^2
- (C) 16 cm^2
- (D) 11 cm^2

21. The instrument fitted in a vehicle that shows its speed at every moment is the:

- (A) speedometer
- (B) thermometer
- (C) odometer
- (D) barometer

22. Written as a decimal, the fraction $\frac{3}{5}$ is:

- (A) 5.3
- (B) 0.6
- (C) 3.5
- (D) 0.35

23. In water and air, heat is carried by the actual movement of the heated material itself. This is called:

- (A) freezing
- (B) conduction
- (C) radiation
- (D) convection

24. A rectangle is 8 cm long and 3 cm broad. Its perimeter is:

- (A) 11 cm
- (B) 22 cm
- (C) 24 cm
- (D) 16 cm

25. The instrument in a vehicle that records the total distance it has travelled is the:

- (A) compass
- (B) speedometer
- (C) odometer
- (D) stopwatch

26. Which fraction is greater, $\frac{2}{3}$ or $\frac{3}{5}$?

- (A) $\frac{2}{3}$
- (B) neither can be compared
- (C) $\frac{3}{5}$
- (D) they are equal

27. The Sun's heat reaches the Earth across empty space, where there is no air, by the process of:

- (A) conduction
- (B) radiation
- (C) convection
- (D) evaporation

28. The area of a square of side 7 m is:

- (A) 14 m^2
- (B) 77 m^2
- (C) 28 m^2
- (D) 49 m^2

29. The to-and-fro (back-and-forth) movement of a swinging pendulum is called an:

- (A) explosion
- (B) rotation
- (C) oscillation
- (D) translation

30. The improper fraction $\frac{7}{4}$ written as a mixed number is:

- (A) $3\frac{1}{4}$
- (B) $1\frac{1}{4}$
- (C) $7\frac{1}{4}$
- (D) $1\frac{3}{4}$

31. Substances like iron and copper, which allow heat to pass through them readily, are known as:

- (A) conductors of heat
- (B) liquids
- (C) insulators of heat
- (D) gases

32. An equilateral triangle has each side equal to 6 cm. Its perimeter is:

- (A) 6 cm
- (B) 18 cm
- (C) 36 cm
- (D) 12 cm

33. The time a simple pendulum takes for one complete to-and-fro swing is known as its:

- (A) speed
- (B) time period
- (C) distance
- (D) frequency

34. The reciprocal of the number 5 is:

- (A) 10
- (B) 0.5
- (C) $\frac{1}{5}$
- (D) 5

35. Materials such as wool, wood and plastic, which do not let heat pass through easily, are called:

- (A) magnets
- (B) metals
- (C) insulators of heat
- (D) conductors of heat

36. The area of a triangle with base 10 cm and height 4 cm is:

- (A) 14 cm^2
- (B) 20 cm
- (C) 40 cm^2
- (D) 20 cm^2

37. The basic SI unit of time is the:

- (A) metre
- (B) kilogram
- (C) hour
- (D) second

38. One-third of the number 90 works out to:

- (A) 27
- (B) 3
- (C) 30
- (D) 60

39. On a hot sunny day, light-coloured clothes keep us more comfortable than dark clothes because light colours:

- (A) make their own cold air
- (B) reflect more heat and absorb less
- (C) stop sweat completely
- (D) absorb more heat and reflect less

40. Using $\pi = 22/7$, the circumference of a circle of radius 7 cm is:

- (A) 44 cm
- (B) 22 cm
- (C) 154 cm
- (D) 14 cm

41. A train travels 150 km in 3 hours at a steady pace. Its speed is:

- (A) 450 km/h
- (B) 153 km/h
- (C) 75 km/h
- (D) 50 km/h

42. The decimal 0.75 written as a fraction in its lowest terms is:

- (A) $1/4$
- (B) $75/10$
- (C) $7/5$
- (D) $3/4$

43. Woollen clothes keep us warm in winter mainly because:

- (A) wool is a very good conductor
- (B) wool is always hot to touch
- (C) wool makes heat by itself
- (D) they trap air, which is a poor conductor of heat

44. Using $\pi = 22/7$, the area of a circle of radius 7 cm is:

- (A) 154 cm^2
- (B) 49 cm^2
- (C) 44 cm^2
- (D) 22 cm^2

45. Which is faster: a scooter moving at 10 m/s or a cycle moving at 30 km/h? (Hint: 10 m/s = 36 km/h.)
- (A) the scooter at 10 m/s
 - (B) the cycle at 30 km/h
 - (C) you cannot compare them at all
 - (D) both are exactly equal
46. The value of 5.6×10 is:
- (A) 560
 - (B) 5.60
 - (C) 0.56
 - (D) 56
47. For an experiment needing temperatures well above 42°C , a student should use a:
- (A) laboratory thermometer
 - (B) barometer
 - (C) clinical thermometer
 - (D) measuring tape
48. A square garden has a side of 25 m. The length of fencing needed to go once around it is:
- (A) 29 m
 - (B) 100 m
 - (C) 50 m
 - (D) 625 m
49. On a distance-time graph, a straight line slanting upward shows that the body is moving with:
- (A) changing speed
 - (B) uniform (steady) speed
 - (C) backward motion
 - (D) no motion
50. The value of $4.5 / 10$ is:
- (A) 4.50
 - (B) 0.045
 - (C) 45
 - (D) 0.45
51. Why can a clinical thermometer NOT be used to measure the temperature of boiling water?
- (A) its scale only goes up to about 42°C , far below boiling
 - (B) it can only measure cold things
 - (C) water has no temperature
 - (D) it measures speed, not heat

52. The area of a parallelogram with base 6 cm and height 5 cm is:
- (A) 11 cm²
 - (B) 30 cm²
 - (C) 22 cm²
 - (D) 15 cm²
53. On a distance-time graph, a flat horizontal line means the body is:
- (A) moving very fast
 - (B) speeding up
 - (C) moving at steady speed
 - (D) at rest (not moving)
54. The sum $2\frac{1}{2} + 1\frac{1}{2}$ equals:
- (A) $4\frac{1}{2}$
 - (B) 3
 - (C) $3\frac{1}{2}$
 - (D) 4
55. A small bend, called a kink, in the tube of a clinical thermometer is there to:
- (A) stop the mercury from slipping back before you read it
 - (B) measure two temperatures at once
 - (C) make the thermometer look attractive
 - (D) let the mercury fall back quickly
56. A rectangular field is 50 m long and 30 m broad. Its area is:
- (A) 80 m²
 - (B) 1500 m²
 - (C) 150 m²
 - (D) 160 m²
57. A cyclist rides at a steady 40 km/h for 3 hours. The distance covered is:
- (A) 13 km
 - (B) 120 km
 - (C) 80 km
 - (D) 43 km
58. The value of 1.2×5 is:
- (A) 6
 - (B) 6.5
 - (C) 1.7
 - (D) 60
59. In a pan of water heated from below, the hot water at the bottom:
- (A) sinks straight to the bottom
 - (B) stays exactly where it is
 - (C) turns instantly to ice
 - (D) rises upward while cooler water sinks down

60. A square plot of side 12 m is to be fenced at ₹10 per metre. The total cost of fencing is:
- (A) ₹1440
 - (B) ₹480
 - (C) ₹48
 - (D) ₹120
61. A car must travel 100 km at a steady speed of 50 km/h. The time it will take is:
- (A) 150 hours
 - (B) 2 hours
 - (C) 50 hours
 - (D) 5 hours
62. The sum $0.1 + 0.01$ equals:
- (A) 0.11
 - (B) 0.11 is wrong, it is 1.1
 - (C) 0.02
 - (D) 0.2
63. Dark-coloured surfaces, compared with light-coloured ones, are:
- (A) better absorbers of heat
 - (B) better reflectors of heat
 - (C) always colder than light ones
 - (D) unable to take in any heat
64. A square has a side of 4.5 cm. Its perimeter is:
- (A) 20.25 cm
 - (B) 18 cm
 - (C) 9 cm
 - (D) 16 cm
65. A pendulum takes 20 seconds to complete 10 full oscillations. Its time period is:
- (A) 200 seconds
 - (B) 2 seconds
 - (C) 10 seconds
 - (D) 20 seconds
66. Reduced to its lowest terms, the fraction $\frac{6}{8}$ becomes:
- (A) $\frac{2}{4}$
 - (B) $\frac{12}{16}$
 - (C) $\frac{3}{4}$
 - (D) $\frac{6}{8}$ is already lowest
67. The handles of cooking pans are often made of plastic or wood because these materials are:
- (A) good conductors, so the handle heats quickly
 - (B) magnets that hold the pan together
 - (C) poor conductors, so the handle stays cool to hold
 - (D) able to make the pan hotter

68. A square has a side of 0.5 m. Its area is:

- (A) 0.25 m^2
- (B) 0.5 m^2
- (C) 1 m^2
- (D) 2 m^2

69. Among these animals, which one moves with the highest speed?

- (A) a man running at 10 m/s
- (B) a tortoise at 0.5 m/s
- (C) a cheetah at 30 m/s
- (D) a flying bird at 2 m/s

70. The value of $1 - \frac{3}{8}$ is:

- (A) $\frac{2}{8}$
- (B) $\frac{1}{8}$
- (C) $\frac{5}{8}$
- (D) $\frac{3}{8}$

71. During the day at the seaside, a cool breeze blows from the sea toward the land because:

- (A) wind has nothing to do with heat
- (B) the sea heats up faster than the land
- (C) the sea is always hotter than the land
- (D) land heats up faster than the sea, and cooler air moves in from the sea

72. A floor is 4 m long and 3 m wide. How many square tiles of side 1 m are needed to cover it exactly?

- (A) 24 tiles
- (B) 12 tiles
- (C) 14 tiles
- (D) 7 tiles

73. Two pendulums swing side by side. The one with the longer time period is the one that:

- (A) swings more slowly
- (B) covers more distance per swing
- (C) swings more quickly
- (D) does not swing at all

74. The sum $\frac{2}{5} + \frac{1}{5}$ equals:

- (A) $\frac{3}{25}$
- (B) $\frac{3}{5}$
- (C) $\frac{2}{5}$
- (D) $\frac{3}{10}$

- 75.** Two thin blankets together often keep you warmer than one thick blanket because:
- (A) two blankets weigh more, and weight makes heat
 - (B) a layer of air trapped between them is a poor conductor of heat
 - (C) air is a very good conductor of heat
 - (D) thin blankets are naturally hotter
- 76.** A rectangle is 12 cm long and 8 cm broad. Its perimeter is:
- (A) 20 cm
 - (B) 40 cm
 - (C) 24 cm
 - (D) 96 cm
- 77.** When a body's speed changes during a journey, the single speed that describes the whole trip is the:
- (A) average speed
 - (B) starting speed
 - (C) top speed
 - (D) zero speed
- 78.** What is 0.25 of 200?
- (A) 50
 - (B) 25
 - (C) 75
 - (D) 100
- 79.** A steel spoon left standing in a cup of hot tea slowly becomes hot to touch. This shows heat travelling by:
- (A) evaporation
 - (B) convection
 - (C) conduction
 - (D) radiation
- 80.** A rectangle has an area of 24 cm^2 and a length of 6 cm. Its breadth is:
- (A) 4 cm
 - (B) 30 cm
 - (C) 144 cm
 - (D) 18 cm
- 81.** Two towns are 240 km apart. A bus travelling steadily at 60 km/h will cover this in:
- (A) 14400 hours
 - (B) 3 hours
 - (C) 180 hours
 - (D) 4 hours

82. Written as a decimal, the fraction $\frac{9}{10}$ is:

- (A) 9.0
- (B) 0.9
- (C) 1.9
- (D) 0.09

83. An electric room heater warms the air in a room, and the warm air spreads around the room mainly by:

- (A) reflection
- (B) freezing
- (C) conduction
- (D) convection

84. The perimeter of a square is 36 cm. The length of one side is:

- (A) 9 cm
- (B) 36 cm
- (C) 6 cm
- (D) 18 cm

85. A boy walks 1.5 km in 30 minutes (half an hour). His speed is:

- (A) 0.5 km/h
- (B) 1.5 km/h
- (C) 3 km/h
- (D) 45 km/h

86. Which statement about 0.3 and 0.30 is correct?

- (A) 0.3 is bigger than 0.30
- (B) 0.30 is ten times 0.3
- (C) 0.30 is bigger than 0.3
- (D) they are equal in value

87. When a substance is heated, its temperature usually:

- (A) falls
- (B) stays exactly the same forever
- (C) disappears completely
- (D) rises

88. A piece of wire 24 cm long is bent to form a square. The length of each side of the square is:

- (A) 24 cm
- (B) 12 cm
- (C) 6 cm
- (D) 96 cm

89. The Earth going around the Sun once every year, again and again, is an example of:
- (A) non-periodic motion
 - (B) random motion
 - (C) periodic motion
 - (D) straight-line motion
90. The value of $\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}$ is:
- (A) $\frac{1}{8}$
 - (B) $\frac{1}{2}$
 - (C) $\frac{1}{6}$
 - (D) $\frac{3}{6}$
91. Why do many cooking pots have shiny, polished outer surfaces?
- (A) shiny surfaces cool the food quickly
 - (B) shiny surfaces absorb heat the fastest
 - (C) shiny surfaces reflect heat and lose it slowly, keeping food warm
 - (D) shiny surfaces make the food tastier
92. A rectangular solar panel is 2 m long and 1 m wide. The area that catches sunlight is:
- (A) 3 m^2
 - (B) 1 m^2
 - (C) 2 m^2
 - (D) 6 m^2
93. On the same distance-time graph, the line that is steeper belongs to the body moving at:
- (A) no motion at all
 - (B) the greater speed
 - (C) exactly the same speed
 - (D) the smaller speed
94. A pendulum has a time period of 0.5 seconds. In 2 seconds, how many complete oscillations does it make?
- (A) 2
 - (B) 0.5
 - (C) 1
 - (D) 4
95. Air, compared with metals, is generally a:
- (A) magnet for heat
 - (B) the same as metal for heat
 - (C) very good conductor of heat
 - (D) poor conductor of heat

96. A rectangular notice board is 1.5 m long and 2 m wide. Its area is:

- (A) 7 m²
- (B) 1.5 m²
- (C) 3.5 m²
- (D) 3 m²

97. Long ago, before clocks, people measured time during the day using the moving shadow on a:

- (A) compass
- (B) speedometer
- (C) sundial
- (D) thermometer

98. A car covers $\frac{3}{4}$ of a 200 km journey. How many kilometres has it travelled?

- (A) 75 km
- (B) 50 km
- (C) 150 km
- (D) 100 km

99. To find out whether a child has a fever, the most useful thing to measure is the body's:

- (A) speed
- (B) temperature
- (C) height
- (D) weight

100. A path runs around the inside edge of a square park of side 20 m. The total distance once around the park is:

- (A) 40 m
- (B) 24 m
- (C) 80 m
- (D) 400 m